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High-content high-resolution microscopy and deep learning assisted analysis reveals host and bacterial heterogeneity during *Shigella* infection

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This manuscript describes an AI-automated microscopy-based approach to characterize both bacterial and host cell responses associated with *Shigella* infection of epithelial cells. The methodology is **compelling** and should be helpful for investigators studying a variety of intracellular pathogens. The authors have acquired **important** findings regarding host and bacterial responses in the context of infection, which should be followed up with further mechanistic-based studies.

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Abstract

Shigella flexneri is a Gram-negative bacterial pathogen and causative agent of bacillary dysentery. *S. flexneri* is closely related to *Escherichia coli* but harbors a virulence plasmid that encodes a Type III Secretion System (T3SS) required for host cell invasion. Widely recognized as a paradigm for research in cellular microbiology, *S. flexneri* has emerged as important to study mechanisms of cell-autonomous immunity, including septin cage entrapment. Here we use high-content high-resolution microscopy to monitor the dynamic and heterogeneous *S. flexneri* infection process by assessing multiple host and bacterial parameters (DNA replication, protein translation, T3SS activity). In the case of infected host cells, we report a reduction in DNA and protein synthesis together with morphological changes that suggest *S. flexneri* can induce cell-cycle arrest. We developed an artificial intelligence image analysis approach using Convolutional Neural Networks to reliably quantify, in an automated and unbiased manner, the recruitment of SEPT7 to intracellular bacteria. We discover that heterogeneous SEPT7 assemblies are recruited to bacteria with increased T3SS activation. Our automated microscopy workflow is useful to illuminate diverse host and bacterial interactions at the single-cell and population level, and to fully characterise the intracellular microenvironment controlling the *S. flexneri* infection process.

Introduction

Shigella flexneri is a Gram-negative bacterium and the etiologic agent of bacillary dysentery, an intestinal infectious disease that kills more than 150 thousand people worldwide per year, disproportionately affecting children under 5 years old from low- and middle-income countries (Global Burden of Disease Collaborative Network, 2020 [\[1\]](#)). After the ingestion of a small number of bacteria (10-100), *S. flexneri* rapidly colonises the human intestinal epithelium. For this, *S. flexneri* uses a Type 3 Secretion System (T3SS) that injects bacterial effectors into the host cell to modulate key aspects of the infection process and to evade innate immune defences by the host (Schnupf & Sansonetti, 2019 [\[2\]](#)). Following uptake by host cells, *S. flexneri* breaks its vacuolar compartment to access the cytosol (Ray et al., 2010 [\[3\]](#)). Here, the virulence factor IcsA enables *S. flexneri* to hijack host actin and polymerise comet tails that render bacteria motile to disseminate to neighbouring cells and spread in the tissue (Bernardini et al., 1989 [\[4\]](#); Egile et al., 1999 [\[5\]](#)).

Septins are highly-conserved GTP-binding proteins that form heteromeric complexes (hexamers, octamers) that assemble into non-polar filaments, bundles and rings (Martins et al., 2023 [\[6\]](#); Soroosh et al., 2021 [\[7\]](#); Szuba et al., 2021 [\[8\]](#); Woods & Gladfelter, 2021 [\[9\]](#)). Septins are involved in many physiological cellular processes (eg. cytokinesis, cilia formation), and play key roles during infection (Mostowy & Cossart, 2012 [\[10\]](#); Robertin & Mostowy, 2020 [\[11\]](#); Spiliotis & Nakos, 2021 [\[12\]](#)). Septins form collar-like structures at the plasma membrane during bacterial invasion (Boddy et al., 2018 [\[13\]](#); Mostowy et al., 2009 [\[14\]](#); Robertin et al., 2023 [\[15\]](#)), and ring-like structures surrounding membrane protrusions in cell-to-cell bacterial spread (Mostowy et al., 2010 [\[16\]](#)). In the cytosol septins are recruited to actin-polymerising bacteria, including *S. flexneri* and *Mycobacterium marinum*, as cage-like structures (Mostowy et al., 2010 [\[16\]](#)). In the case of *S. flexneri*, septins are recruited to μm -scale curvature at bacterial poles and the division site, which are enriched in the negatively charged phospholipid cardiolipin (Krokowski et al., 2018 [\[17\]](#)). Recent work has shown that IcsA promotes the recruitment of septins to *S. flexneri*, and that lipopolysaccharide (LPS) inhibits *S. flexneri*-septin cage entrapment (Lobato-Márquez et al., 2021 [\[18\]](#); Mostowy et al., 2010 [\[16\]](#)). Septin cages prevent bacterial actin tail motility and therefore are considered as an antibacterial mechanism of host defence (Krokowski et al., 2018 [\[17\]](#); Mostowy et al., 2010 [\[16\]](#)).

Intracellular infection is usually portrayed as a uniform, synchronised and highly orchestrated process. However, infection assays performed in controlled laboratory environments with isogenic populations display a high level of complexity (Avraham et al., 2015 [\[19\]](#); Toniolo et al., 2021 [\[20\]](#)). It is particularly unknown how different subsets of bacteria (eg. motile vs. non motile bacteria) can colonise distinct intracellular microenvironments and become targeted by multiple cell-autonomous immunity mechanisms (eg. septin cages, autophagy). It has been proposed that this diversity of intracellular niches impacts bacterial physiology (Day et al., 2023 [\[21\]](#); Gutierrez & Enninga, 2022 [\[22\]](#); Santucci et al., 2021 [\[23\]](#)). For example, it may affect individual bacterial ability to acquire nutrients, to be exposed to varying antibiotic concentrations, or to elicit different responses and signalling pathways within the host cell. While the concept of phenotypic heterogeneity in prokaryotic clonal populations is well established (Ackermann, 2015 [\[24\]](#)), the concept is emerging also in eukaryotic cells where non genetic differences may for instance explain individual cell sensitivity and responses by cell-autonomous immunity (McDonald & Dedhar, 2024 [\[25\]](#)). In addition, previous high-content studies coupled with mathematical modelling have shown that host cell context (including local cell density, lipid composition at the plasma membrane, or previous infection status) affects their susceptibility to viral and bacterial infection (Snijder et al., 2009 [\[26\]](#); Voznica et al., 2018 [\[27\]](#)). It is therefore paramount to study host and bacterial cells at the single-cell level, to assess the individual responses that collectively contribute to the infection process.

Here, to dissect the complexity of *S. flexneri* infection in epithelial cells, we use high-content high-resolution microscopy coupled to automated image analysis. High-content microscopy enables the automated acquisition of thousands of microscopy images and therefore is useful to capture the heterogeneity of individual host and bacterial cells during the infection process for quantitative analysis (Aylan et al., 2023 [↗](#); Brodin & Christophe, 2011 [↗](#); Deboosere et al., 2021 [↗](#); Dramé et al., 2023 [↗](#); Fisch et al., 2019a [↗](#); Lensen et al., 2023 [↗](#); Pylkkö et al., 2021 [↗](#)). We analyse multiple parameters, such as morphological host cell features, the *de novo* synthesis of DNA and proteins in both host and bacterial cells, as well as activation of the bacterial T3SS. To investigate the antimicrobial potential of septin-mediated immunity, we develop a deep-learning based approach to automatically identify septin-*S. flexneri* interactions. Together, we highlight the power of high-content microscopy and artificial intelligence to reveal insights on both host and bacterial cells, and illuminate fundamental biology underlying the *S. flexneri* infection process.

Results

Capturing the heterogeneity of *S. flexneri* infection using high-content microscopy

To understand the complexity and reveal the heterogeneity of *S. flexneri* infection in human epithelial cells, we dissected the infection process using high-content high-resolution microscopy and HeLa cell infection model (**Figure 1A, B** [↗](#), **Figure 1-figure supplement 1A** [↗](#)). We observed a mean of 11.5% of infected cells measured by fluorescence at 3h 40 min post infection (hpi) at MOI 100:1 (**Figure 1C** [↗](#)). These infected cells presented different burdens of bacteria following an exponential decay distribution (**Figure 1D** [↗](#)), reflective of the formation of infection foci (Ortega et al., 2019 [↗](#)). With growing evidence of the cytosol being a non-homogeneous compartment containing concentration gradients, biomolecular condensates, protein complexes and concentrates (van Tartwijk & Kaminski, 2022 [↗](#)), we tested whether *S. flexneri* may be exposed to different intracellular microenvironments. For this, we measured the distance of bacteria to the centroid of cells (**Figure 1E** [↗](#)) and observed a median distance of 8.74 μm . The median distance to the centroid of the cell significantly decreased with increasing bacterial burden (**Figure 1E** [↗](#)), suggesting that the microenvironment of bacteria is diverse and changes as bacteria replicate intracellularly and populate the central parts of host cytosol after uptake at the host cell periphery. To dissect the heterogeneity in host cells, we quantified an increase of 36.9% and 22.3% in the cellular and nuclear area of infected cells, respectively (**Figure 1G, I** [↗](#)), which was dependent on the infection burden (**Figure 1H, J** [↗](#)). In addition, the total amount of Hoechst fluorescence in the cell nucleus increased 29.4% in infected cells (**Figure 1-figure supplement 1B** [↗](#)), indicating either an increase in nuclear volume or altered chromatin condensation. To test if this effect was due to morphological rearrangements during infection, we treated cells with Latrunculin B to inhibit actin polymerisation. While cells treated with Latrunculin B decreased their cellular area (but not their nuclear area) (**Figure 1-figure supplement 1C, D** [↗](#)), the drug did not abrogate the difference in size between infected and uninfected cells. Taken together, these results highlight profound morphological rearrangements during infection that are dependent on the infection burden.

Epithelial cells reduce *de novo* DNA and protein synthesis upon infection

In order to understand the physiological state of enlarged epithelial cells infected with *S. flexneri*, we monitored host DNA replication and protein translation using Click-chemistry. Using a 1 h pulse of the clickable thymidine analogue Edu, we specifically visualized and quantified *de novo* DNA synthesis in the population of epithelial cells with high-content microscopy (**Figure 2A** [↗](#), **Figure 2-figure supplement 1** [↗](#)). We identified the subpopulation of cells in S-phase by their high

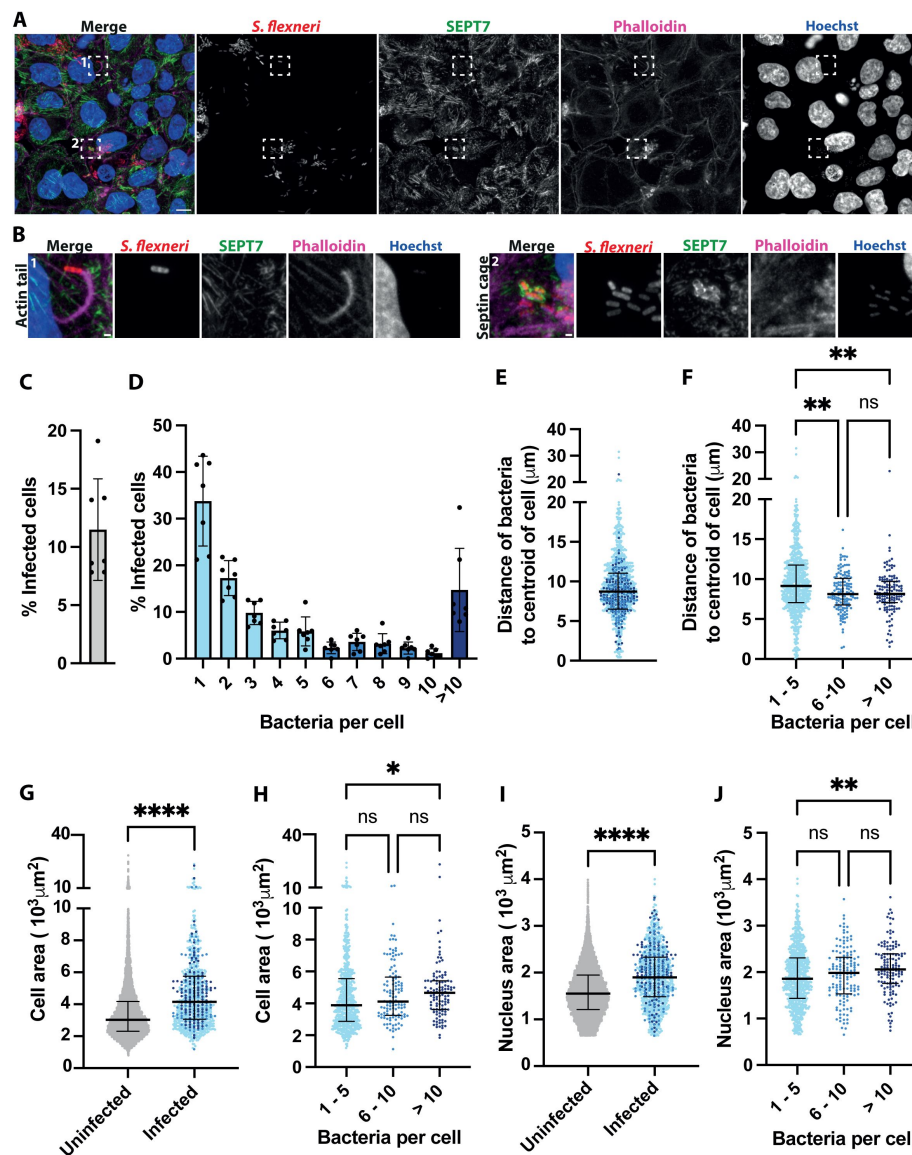


Figure 1.

High-content imaging captures the heterogeneity of *S. flexneri* infection.

(A) Representative high-resolution microscopy image of HeLa cells infected with *S. flexneri* expressing mCherry acquired with the high-content microscope ZEISS CellDiscoverer7. Scale bar, 10 μm. (B) Enlarged view of insets in A, highlighting a *S. flexneri* bacteria polymerising an actin tail (1) and entrapped in a septin cage (2), scale bar, 1 μm. (C) Percentage of infected cells at 3 h.p.i. of 7 independent experiments. Graph represents mean = 11.49% ± SD = 4.36%. (D) Percentage of infected cells harbouring different bacterial doses, graph represents mean ± SD. (E) Distance of the bacteria to the centroid of the host cell. Graph represents the median (8.74 μm) and interquartile range (6.51 – 11.05 μm). Values from n = 4990 cells from 7 independent experiments. (F) Distance of individual bacteria to the centroid of the host cell depending on its bacterial load. Graph represents the median and interquartile range. Kruskal-Wallis test and Dunn's multiple comparisons test. (G) Area of cells depending on infection. Graph represents the median area (uninfected: 3031 μm², infected: 4148 μm²) and interquartile range (uninfected: 2308 – 4169 μm², infected: 3062 – 5752 μm²). Values from n = 9469 uninfected cells and n = 1031 infected cells from 7 independent experiments. (H) Area of infected cells depending on its bacterial load. Graph represents the median and interquartile range. Kruskal-Wallis test and Dunn's multiple comparisons test. (I) Nuclear area of cells depending on infection. Graph represents the median area (uninfected: 1551 μm², infected: 1897 μm²) and interquartile range (uninfected: 1209 – 1950 μm², infected: 1486 – 2333 μm²). Values from n = 9469 uninfected cells and n = 1031 infected cells from 7 independent experiments. (J) Nuclear area of infected cells depending on its bacterial load. Graph represents the median and interquartile range. Kruskal-Wallis test and Dunn's multiple comparisons test.

level of Edu incorporation. Consistent with this, this subpopulation of cells was fully abrogated in the presence of the S-phase blocker Aphidicolin (Pedrali-Noy et al., 1980) (Figure 2-figure supplement 2A, B). Approximately one third of uninfected cells (34.45%) were found in S-phase (Figure 2B, C), in agreement with S-phase lasting around one third of the cell cycle (~8h out of ~22h). Infected cells presented a similar proportion of cells in S-phase (35.89%), indicating that *S. flexneri* infection did not block transition of cells from G1 to S at the timepoint tested. We then assessed the amount of *de novo* DNA synthesis incorporated in cells in S-phase during infection. Surprisingly, we observed a reduction of Edu incorporation in infected cells (Figure 2D), which was dependent on the bacterial burden (Figure 2E). These results indicate that *S. flexneri* infection slows down DNA synthesis of epithelial cells.

To assess protein translation, we used a 1h pulse of the clickable methionine analogue AHA, which specifically labels *de novo* synthesized proteins as AHA incorporation is reduced after treatment with the inhibitor of translational elongation Cycloheximide (Ennis & Lubin, 1964) (Figure 2-figure supplement 1C, D). During infection, we observed a reduction of protein translation in infected cells compared to non-infected cells (Figure 2F,G), which was also dependent on the bacterial burden (Figure 2H).

Collectively, the cell morphological rearrangements together with the reduction of *de novo* DNA and protein synthesis in infected epithelial cells suggests that *S. flexneri* induces an arrest in the host cell cycle.

Design of a deep learning pipeline to automatically identify bacteria interacting with SEPT7

S. flexneri have been shown to interact with septins during infection. These interactions include septin ring-like structures surrounding phagocytic cups, autophagosomes, actin tails and protrusions, all being morphologically diverse structures that are normally scored by hand (Krokowski et al., 2018; Lobato-Márquez et al., 2023; Mostowy et al., 2010). To study septin-associated bacteria using high-content microscopy, we designed a tailored analysis pipeline for the automatic and unbiased identification of SEPT7-bacteria interactions (Figure 3). SEPT7 is a core component of septin hexamers and octamers, and is historically used to identify septin structures (Mostowy et al., 2010). The analysis pipeline includes two deep learning models based on Convolutional Neural Networks (CNN), which are effective in solving classification tasks in computer vision (Krizhevsky et al., n.d.; Lecun et al., 2015; Nielsen, 2015). As septins can surround individual and also multiple bacteria simultaneously (Figure 4-figure supplement 1A), we focused on individual bacteria. For this, we trained the first classification model to discriminate between “single” bacteria and “clumps”. The second classification model was designed to assign those single bacteria into “SEPT7-positive structures” or alternatively “negatives”, according to the morphology of SEPT7 recruitment around them.

The first classification model (discerning “single” versus “clumped” bacteria) together with its training and performance metrics is summarized in Figure 4. A large dataset of annotated bacteria (Figure 4A, B, Figure 4-figure supplement 1B) was used to train a Sequential model with a typical CNN architecture that was experimentally tested and optimised accordingly for best results (Figure 4C). This model provided a good fit of the training data (high accuracy, low loss), with similar predictive power for the validation dataset (Figure 4D, E). Performance of the model was further assessed with a confusion matrix where a separate annotated test dataset was challenged and correct number of predictions scored, showing that 96% of SEPT7 structures and 83% of clumps were correctly identified (Figure 4F). Finally, effective performance of the model was ensured with high values achieved in additional metrics testing for type I and II errors (precision, recall, F1-score) (Figure 4G).

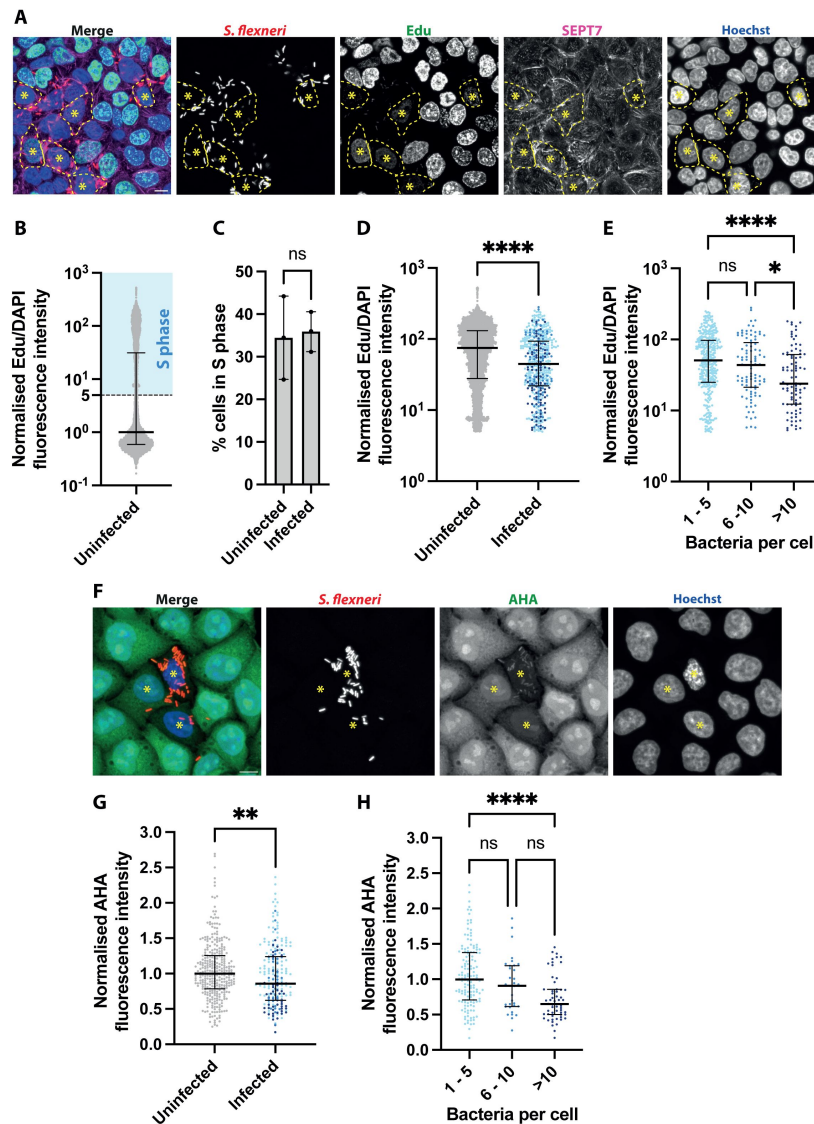


Figure 2

***S. flexneri* infection reduces host DNA synthesis and host protein synthesis.**

(A) Representative microscopy image of HeLa cells infected with *S. flexneri* expressing mCherry at 3 h.p.i. after 1h incubation with Click-IT Edu. Asterisks label the nucleus of infected cells with reduced Edu incorporation. Contour of selected cells is delimited with a dashed line. Scale bar, 10 μ m. (B) Edu acquisition in uninfected cells is bimodal. Graph represents the normalised median of the ratio between the fluorescence intensity of Edu and Hoechst (1), and interquartile range (0.5924 – 31.22). Values from $n = 8941$ cells from 3 independent experiments. An arbitrary threshold of 5 is selected to define cells in S-phase. (C) Frequency of cells in S-phase (as defined in B) depending on infection. Graph represents mean $\pm$ SD (uninfected: $34.45\% \pm 9.78\%$, infected: $35.89\% \pm 4.69\%$). Student's *t*-test. (D) Edu acquisition in cells in S-phase depending on infection. Graph represents the normalised median of the ratio between the fluorescence intensity of Edu and Hoechst (uninfected: 75.14, infected: 44.73), and interquartile range (uninfected: 27.89 – 131.0, infected: 22.05 – 93.36), from 3065 uninfected and 552 infected cells. Mann-Whitney U test. (E) Edu acquisition in cells in S-phase depending on bacterial load. Graph represents the median and interquartile range. Kruskal-Wallis test and Dunn's multiple comparisons test. (F) Representative microscopy image of HeLa cells infected with *S. flexneri* expressing mCherry at 3 h.p.i. after 1h incubation with Click-IT AHA. Asterisks label the infected cells with reduced AHA incorporation. Scale bar, 10 μ m. (G). AHA acquisition in host cells depending on infection. Graph represents the normalised median of AHA fluorescence intensity (uninfected: 1, infected: 0.86), and interquartile range (uninfected: 0.7850 – 2.694, infected: 0.6207 – 1.241), from 36 uninfected and 233 infected cells. Mann-Whitney U test. (H) AHA acquisition in infected cells depending on bacterial load. Graph represents the median and interquartile range. Kruskal-Wallis test and Dunn's multiple comparisons test.

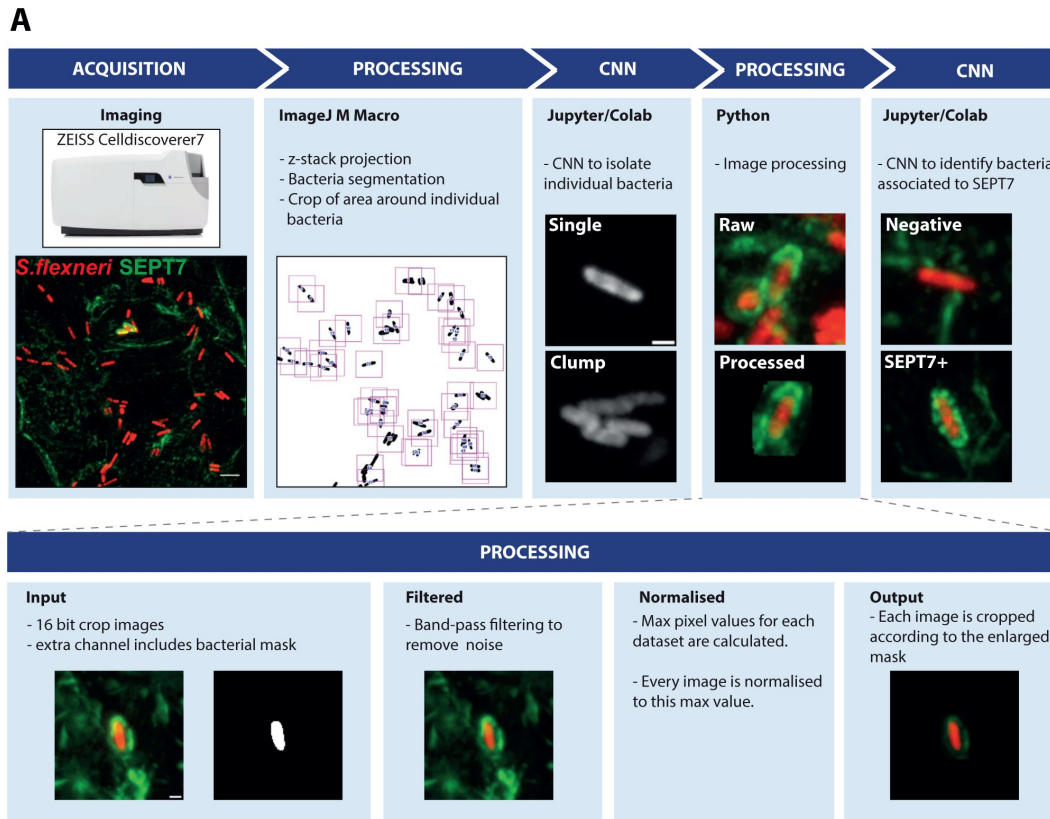


Figure 3.

Microscopy pipeline to automatically identify SEPT7-*S. flexneri* interactions.

(A) Diagram representing the steps involved in the imaging and analysis pipeline. After infection, fixation and staining, high number of images are automatically acquired using the high-content microscope. Scale bar, 10 μm . These are then processed to segment bacteria based on their fluorescence, so that a square field containing the bacteria in the centre is cropped and saved for subsequent analysis. A deep learning model based on CNN is applied using Jupiter notebooks and the Python Keras library hosted on Colab to sort individual, isolated bacteria. Several processing steps are applied using Python to remove noise (band-pass filtering by difference of Gaussian and mean filters), normalise the data across datasets for comparison, and remove signal away from the bacteria which is irrelevant for the identification of SEPT7 assemblies. Scale bar, 1 μm . Finally, a similar second deep learning model based on CNN is applied to identify SEPT7 interacting with bacteria.

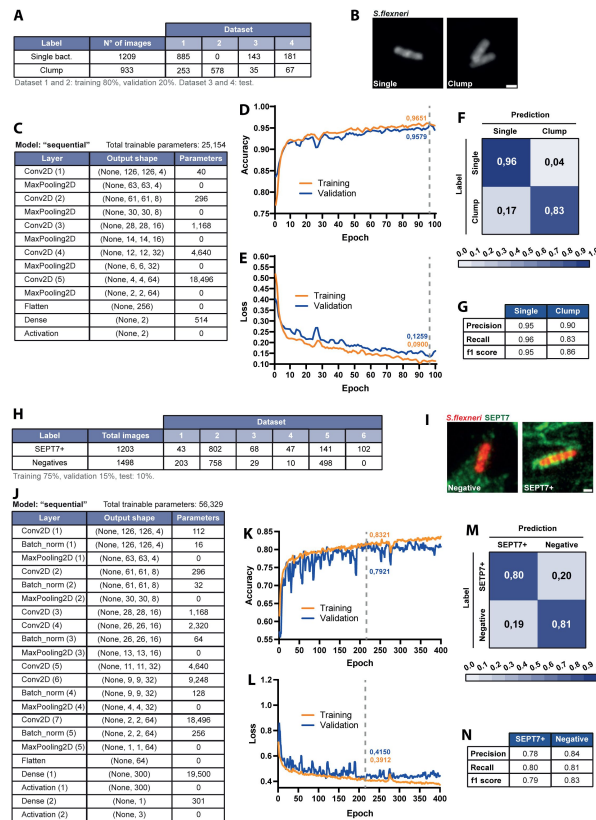


Figure 4.

Training of a deep learning algorithm to identify single isolated bacteria cells and SEPT7 associated *S. flexneri*.

(A) Table summarizing the annotated dataset used for the training process. The labels of the two classification classes used were "Single bacteria" and "Clump". The annotated images were randomly split into two groups, one used as training dataset (comprising 80% of the images) and another used as validation dataset (with the remaining 20%). A separate annotated dataset was used for testing. (B) Representative Airyscan images of bacteria used for each class. 2D max projections of Z-stacks were used for training. Scale bar, 1 μ m. (C) Architecture of the deep learning algorithm used for training. Each row describes the characteristics of the sequential transformation steps applied. Conv2D stands for 2D Convolution. (D, E) Accuracy and Loss as metrics to represent the training process over subsequent epochs (entire passing of the training data through the algorithm). Accuracy increases and loss decreases for both training and validation datasets, indicating a good fit of the model to the data. Vertical dashed grey line indicates Early Stopping, or epoch value for the minimum validation Loss. (F) Confusion matrix performed on an annotated test dataset not used previously for the training or validation process, indicating the percentage of predictions that were correct or wrong for each class. (G) Precision, recall and f1 score as metrics that summarize the performance of the model or classifier. (H) Table summarizing the annotated dataset used for the training process. The labels of the two classification classes used were "Septin" and "Negative". The annotated images were randomly split into three groups, one used as training dataset (comprising 75% of the images), another used as validation dataset (with 15%) and a last one used as test dataset (with 10%). Due to the data being very imbalanced (15% natural frequency of SEPT7 associated bacteria), the images comprised in the Negative class were under-sampled as indicated in the table. (I) Representative Airyscan images of bacteria used for each class. 2D max projections of Z-stacks were used for training. Scale bar, 1 μ m. (J) Architecture of the deep learning algorithm used for training. Each row describes the characteristics of the sequential transformation steps applied. Conv2D stands for 2D Convolution, Batch_norm stands for Batch normalisation. (K, L) Accuracy and Loss as metrics to represent the training process over subsequent epochs. Accuracy increases and loss decreases for both training and validation datasets, indicating a good fit of the model to the data. Vertical dashed grey line indicates Early Stopping, or epoch value for the minimum validation Loss. (M) Confusion matrix performed on the test dataset not used previously for the training or validation process, indicating the percentage of predictions that were correct or wrong for each class. (N) Precision, recall and f1 score as metrics that summarize the performance of the model or classifier.

The second classification task (identification of SEPT7 “positive” vs “negative” bacteria) was more challenging due to their intrinsic heterogeneity plus the presence of ubiquitous SEPT7 filaments that populate the vicinity of bacteria (but do not associate with bacteria per se). To address these challenges, we increased the amount of data and complexity of the second classification model (**Figure 4H-J**, **Figure 4-figure supplement 1C**) compared to the first (**Figure 4A-C**). This included more than 1200 images of *S. flexneri* associated to SEPT7, both from infections with wild-type bacteria or expressing AfaI to increase bacterial invasion. The training process achieved a high accuracy and low loss for the training and validation dataset (**Figure 4K, L**). Correct prediction of manually annotated SEPT7 positive structures was 80% as shown in the confusion matrix (**Figure 4M**), with performance metrics Precision, Recall and F1 score also around 0.8 (**Figure 4N**).

Characterization of the morphological heterogeneity of SEPT7 recruitment to *S. flexneri*

The training of the classification model to identify bacteria associated to SEPT7 required a large dataset of images. We observed diverse morphological patterns of SEPT7 signal around bacteria (**Figure 5 A-C**). We differentiated five frequent subcategories from a total of 855 cases. The first corresponds to most cases (66.7%), where SEPT7 forms rings that entrap bacteria transversally (**Figure 5A, B, a**). We also observed two categories where SEPT7 surrounds bacteria more homogeneously either in a tight (18.2%) or loose association (6.1%) (**Figure 5A, B, b-c**). Finally, we observed partial SEPT7 recruitment, with only one (5.0%) or two (4.0%) bacterial poles being targeted (**Figure 5A, B, d-e**). Given this diversity, we combined the entire dataset to summarise the probability of SEPT7 interactions with bacteria (**Figure 5-figure supplement 1**). As bacteria are found in any orientation inside epithelial cells, we included only those instances where the z-projection of the bacterial axis measured more than 2.5µm, ensuring they were positioned horizontally when imaged. Representations shown in **Figure 5-figure supplement 1A, C** and graphs in **Figure 5-figure supplement 1B, D** can be understood as a probability map of the area of influence of SEPT7 when associating with intracellular *S. flexneri*.

SEPT7 is recruited to *S. flexneri* with increased T3SS activity

Septins have previously been shown to recognise growing bacterial cells *in vitro* and during infection of HeLa cells, using pharmacologic and genetic manipulation (Krokowski et al., 2018; Lobato-Márquez et al., 2021). To test the metabolic state of SEPT7-associated *S. flexneri* at 3h 40 min post infection (hpi) we used our high-content imaging and analysis pipeline (**Figure 6**). We measured *S. flexneri* *de novo* synthesis of DNA and proteins using a 1 h pulse Click-IT Edu and Click-IT AHA. Only active bacteria incorporated these analogues, as the antibiotic Nalidixic acid (inhibitor of the bacterial DNA gyrase (Sugino et al., 1977)) and Rifampicin (inhibitor of the bacterial RNA polymerase (Ezekiel & Hutchins, 1968)) significantly reduced their incorporation, respectively (**Figure 6-figure supplement 1A-D, Figure 6-figure supplement 1E,F**). During infection, bacteria associated with SEPT7 showed heterogeneous levels of DNA synthesis and protein translation similar to SEPT7-negative bacteria (**Figure 6A-H**).

S. flexneri pathogenicity relies on its T3SS. To understand effector secretion at the individual bacterial level, we used *S. flexneri* expressing the Transcription-based Secretion Activity Reporter (TSAR) (Campbell-Valois et al., 2014). This reporter comprises a short lived fast-maturing superfolder GFP (GFPmsf2) that is expressed in conditions where the T3SS is active, such as in broth upon addition of congo red (**Figure 7-figure supplement 1A, B**) (Parsot et al., 1995) and during infection (**Figure 7**). We observed heterogeneity in TSAR signal (**Figure 7A,B**). T3SS activation was higher in bacteria found in cells with a low bacterial burden (less than 10 intracellular bacteria) (**Figure 7A,C**). This suggests the highest level of activation occurs in bacterial pioneers that colonise neighbouring cells, in agreement with previous reports (Campbell-Valois et al., 2014). We then assessed T3SS activation in *S. flexneri* associated with SEPT7 using

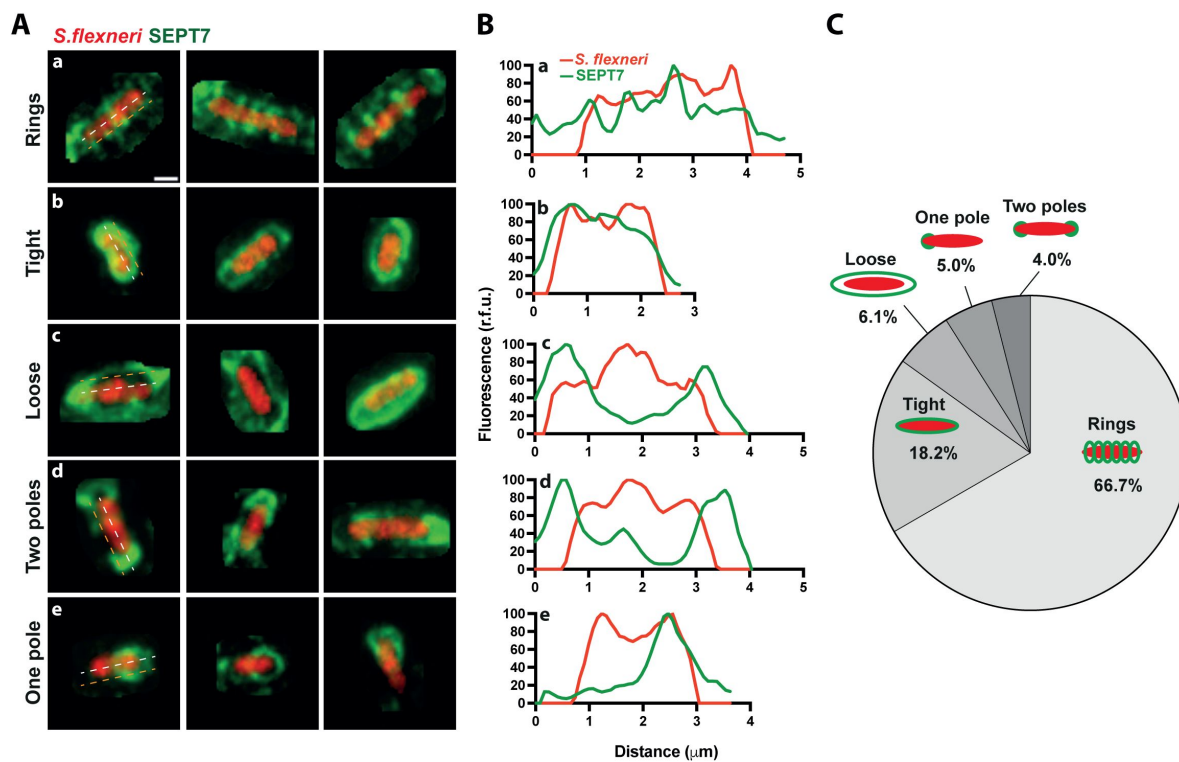


Figure 5.

Septin assemblies associated to *S. flexneri* are heterogeneous.

(A) Selected examples of the training dataset to showcase the diverse morphology of septin assemblies associated to bacteria. Frequent morphologies include (a) ring like structures around the bacteria, (b) smooth tight recruitment around the entire bacterial surface, (c) loose structures more distant from the bacterial surface, (d) septin recruitment to both bacterial poles, (e) septin recruitment to a single bacterial pole. Scale bar, 1 μm . (B) Intensity profiles of the cytosolic bacterial mCherry and the SEPT7 recruitment, as indicated in the white and orange dashed lines, respectively. (C) Pie chart representing the relative frequency of 855 *S. flexneri*-SEPT7 associations depicted in A.

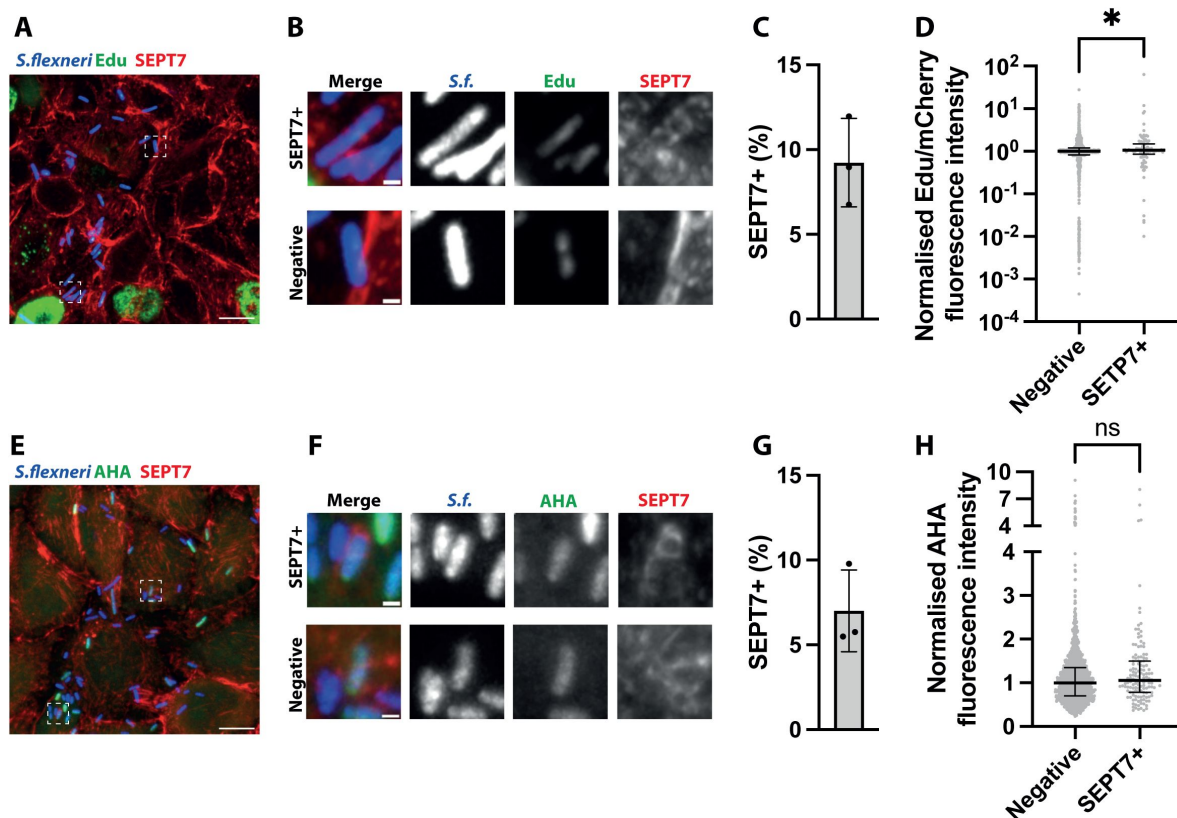


Figure 6.

***S. flexneri* associated with SEPT7 are metabolically active.**

(A) Representative microscopy image of HeLa cells infected with *S. flexneri* expressing mCherry at 3 h.p.i, after 1h incubation with Click-IT Edu. Scale bar 10 μ m. (B) Enlarged view of insets in A, highlighting a *S. flexneri* associated or not to SEPT7 structures. Scale bar, 1 μ m. (C) Frequency of the number of SEPT7-associated bacteria identified using deep learning from experiments in A, from n = 1024 from 3 independent replicates. Graph represents mean $\pm$ SD (9.23% $\pm$ 2.62%). (D) Normalised ratio between Edu signal and mCherry in individual bacteria. Graph represents the median (negatives: 1, septin associated: 1.059), and interquartile range (negatives: 0.8185 – 1.207, septin associated: 0.8573 – 1.487). Mann-Whitney U test. (E) Representative microscopy image of HeLa cells infected with *S. flexneri* expressing mCherry at 3 h.p.i, after 1h incubation with Click-IT AHA. Scale bar 10 μ m. (F) Enlarged view of insets in E, highlighting a *S. flexneri* associated or not to SEPT7 assemblies. Scale bar, 1 μ m. (G) Frequency of the number of SEPT7-associated bacteria identified using deep learning from experiments in E, from n = 1872 bacteria from 3 independent replicates. Graph represents mean $\pm$ SD (7.01% $\pm$ 2.41%). (D) Normalised AHA signal in individual bacteria in A, B. Graph represents the median (negatives: 1, SEPT7 associated: 1.055), and interquartile range (negatives: 0.6986 – 1.345, SEPT7 associated: 0.7837 – 1.497). Mann-Whitney U test.

our deep learning analysis pipeline (**Figure 7D-H**). We detected a significant increase in TSAR-positive bacteria in SEPT7 associated *S. flexneri* compared to SEPT7-negative bacteria (**Figure 7G,H**). These include SEPT7-associated protrusions and cage-like structures (**Figure 7E**, **Figure 7-figure supplement 1C**). These data underscore the antimicrobial role of septins in associating with actively secreting bacteria.

Discussion

Unravelling the intrinsic heterogeneity of host and bacterial cells underlying the infection process is paramount to design effective therapeutic strategies. High-content and high-throughput image analysis are ideal tools to capture this heterogeneity when coupled to automatic phenotypic image analysis (Eliceiri et al., 2012; Mattiazzi Usaj et al., 2016; Pegoraro & Misteli, 2017). Here we use high-content high-resolution microscopy and automated image analysis to quantitatively dissect the heterogeneity of *S. flexneri* in epithelial cells at the single-cell and population level at unprecedented subcellular detail.

We discover that *S. flexneri* induces important morphological and physiological changes in epithelial cells after 3h40min post infection. These include a marked decrease of DNA synthesis in cells in S-phase. Consistent with the ability of bacteria to manipulate the host cell cycle, *S. flexneri* has been shown to induce genotoxic stress (Bergounioux et al., 2012) and block the G2/M phase transition (Iwai et al., 2007; Wang et al., 2019). We also observed a collective shutdown of protein expression, which happened in a bacterial-burden dependent manner. A recent report has shown that the OspC family of *S. flexneri* effectors ADP-ribosylates the host translation initiation factor 3, leading to translational arrest and the formation of stress granules (Zhang et al., 2024). Collectively, these observations suggest that *S. flexneri* infection induces host cell stress and arrests the progression of the cell cycle. In the natural *S. flexneri* niche of the human gut, such an arrest would prevent maturation and shedding of infected epithelial cells, thereby promoting efficient gut colonisation (Iwai et al., 2007).

To understand how bacteria are targeted by cell-autonomous immune defences, such as septin cage entrapment, it is necessary to investigate phenotypic characteristics of targeted bacteria and the heterogeneous recruitment of cell-autonomous immune factors. While traditional image segmentation tools fail to enable this task, rapidly evolving developments in the field of computer vision are democratising the use of deep learning and CNNs for multiple applications (Berg et al., 2019; Hung et al., 2020; Stringer et al., 2020; von Chamier et al., 2021; Weigert et al., 2020) including infection biology (Davidson et al., 2021; Fisch et al., 2019a, 2021; Touquet et al., 2018). However, the analysis of complex datasets still requires a tailored approach, as shown here for bacterial interactions with the septin cytoskeleton (which is ubiquitous in the cell and forms diverse assemblies whose precise identification requires high resolution). We successfully implement a CNN based approach for the automated classification of bacteria associated with SEPT7 assemblies using z-stacks projections, enabling the analysis of thousands of bacterial cells while saving time and preventing human bias. Building upon this work, future studies will improve use of 3D data for CNN re-training instead of z-stack projections, to allow more precise identification of *S. flexneri*-septin interactions, particularly when bacteria are placed perpendicular to the imaging plane.

A large dataset of images depicting *S. flexneri* – SEPT7 interactions was acquired to train the deep learning algorithm and to characterize the diversity of these structures. We observe that the most frequent septin assembly is in the form of rings transversally surrounding the bacterial surface. We also detected alternative morphologies, including homogeneous SEPT7 labelling surrounding the bacterial surface (tight vs loose), as well as the discrete recruitment of SEPT7 to one or two bacterial poles. While previous work has shown that septin cage assembly around *S. flexneri* can start with the discrete recruitment of septins to one or two bacterial poles (Krokowski et al.,

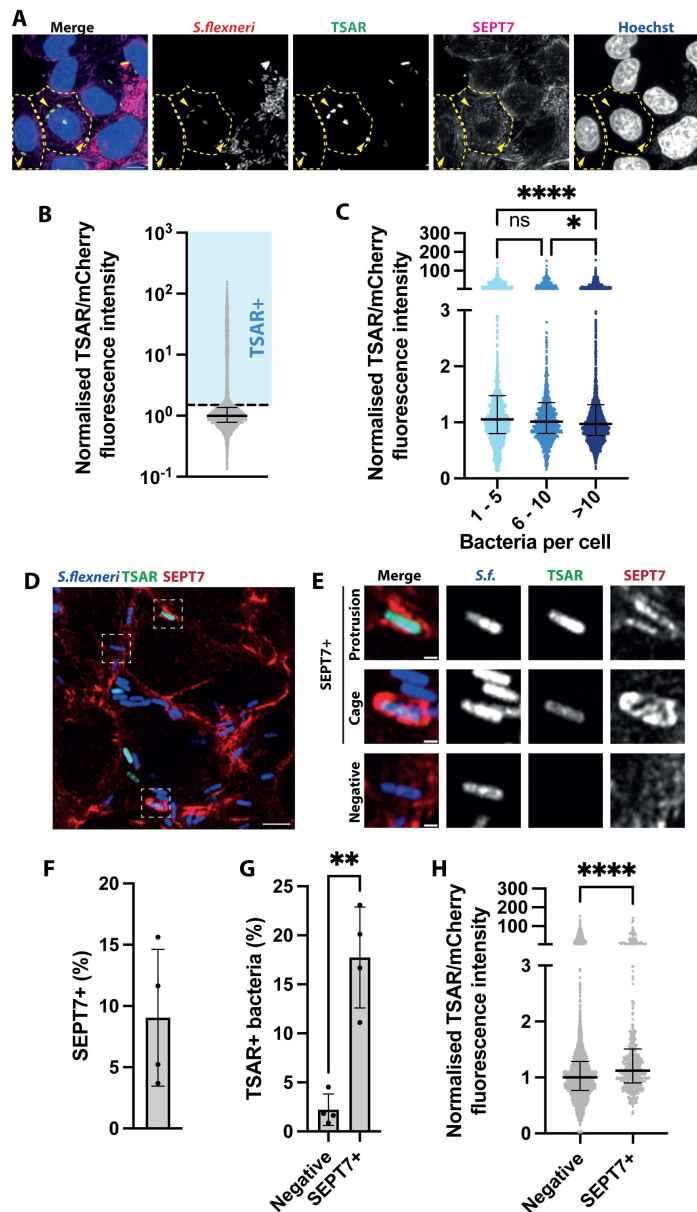


Figure 7.

S. flexneri associated with SEPT7 have an active T3SS.

(A) Representative microscopy image of HeLa cells infected with *S. flexneri* expressing the TSAR reporter at 3 h.p.i. Scale bar 10 μ m. Arroheads highlight TSAR+ bacteria. Selected infected cells contours are delimited with dashed lines. (B) Distribution of TSAR signal intensity in individual bacteria. Graph represents the normalised median of the ratio between TSAR and mCherry (1), and interquartile range (0.7827 – 1.369). Values from $n = 3654$ bacterial cells from 3 independent experiments. An arbitrary threshold of 1.5 is selected to define TSAR+ *S. flexneri*. (C) Intensity of TSAR signal in individual bacteria depending on infection load of host cells. Graph represents the normalised median and interquartile range. Kruskal-Wallis test and Dunn's multiple comparisons test. (D) Representative microscopy image of HeLa cells infected with *S. flexneri* expressing the TSAR reporter at 3 h.p.i. Scale bar 10 μ m. (E) Enlarged view of insets in D, highlighting a *S. flexneri* associated or not to SEPT7 assemblies. Scale bar, 1 μ m. (F) Frequency of the number of SEPT7-associated bacteria identified using deep learning from experiments in F, from $n = 5030$ bacteria from 4 independent replicates. Graph represents mean $\pm$ SD (9.044% $\pm$ 5.582%). (G) Percentage of TSAR+ *S. flexneri* depending on their association to SEPT7. Graph represents mean $\pm$ SD (negatives: 2.214 $\pm$ 1.603, septin associated: 17.74 $\pm$ 5.138). Student's t-test. (H) Normalised ratio between TSAR signal and mCherry in individual bacteria. Graph represents the median (negatives: 1, SEPT7 associated: 1.122), and interquartile range (negatives: 0.7674 – 1.286, SEPT7 associated: 0.9047 – 1.511). Mann-Whitney U test.

2018 [\[link\]](#)), it is not yet known if the different septin assemblies we describe here correspond to different stages or fates of the same structure or whether it marks separate biological events. To fully address this question, follow-up studies should include timelapse analysis of septin cage formation. We use deep learning analysis to interrogate intracellular bacteria and show that *S. flexneri* associated with SEPT7 present heterogeneous DNA and protein synthesis. In addition, we show that bacteria associated with SEPT7 have increased T3SS activity (as compared to bacteria not associated with SEPT7). T3SS activity has been shown to be higher in motile bacteria (Campbell-Valois et al., 2014 [\[link\]](#)) which reinforces the antimicrobial role of septins in counteracting actin tail motility. Altogether, our deep learning tool to analyse septin-bacteria interactions opens the door to future studies on interactions between bacteria and components of cell-autonomous immunity.

Concluding remarks

Heterogeneity in a controlled lab environment is only a small representation of the broader heterogeneity underlying the clinical infection process, considering the genetic background of the patient, interactions with microbiota, etc. In the future, our pipeline can be applied to investigate septin-bacteria interactions among different *Shigella* spp. or clinically relevant strains, as well as other intracellular bacteria (eg. *M. marinum*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*). In addition, our pipeline can be adapted to identify and score other cytoskeletal structures (such as actin tails) or membrane organelles (such as autophagosomes) associated with intracellular bacteria. Ultimately, we envision that our automated microscopy workflows will help to screen pharmacologic or genetic libraries in a multiparametric manner and understand their impact on host and bacterial cells at the single-cell and population level. In this way, novel host and bacterial factors underlying heterogeneity and bacterial pathogenesis could be identified, leading to transformative therapeutic strategies to combat bacterial infection in humans.

Materials and methods

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
gene (<i>Drosophila melanogaster</i>)	nito	GenBank	FLYB:FBgn0027548	
strain, strain background (<i>Escherichia coli</i>)	BL21(DE3)	Sigma-Aldrich	CMC0016	Electrocompetent cells
genetic reagent (<i>D. melanogaster</i>)	MTD-Gal4	Bloomington Drosophila Stock Center	BDSC:31777; FLYB:FBtp001612; RRID:BDSC_31777	FlyBase symbol: P{GAL4-nos.NGT}
cell line (<i>D. melanogaster</i>)	S2	This paper	FLYB:FBtc000181; RRID:CVCL_Z992	Cell line maintained in N. Perrimon lab; FlyBase symbol: S2-DRSC.
cell line (<i>Homo sapiens</i>)	Dermal fibroblast (normal, Adult)	ATCC	PCS-201-012	
transfected construct (<i>M. musculus</i>)	4E-BP1 shRNA #1	VectorBuilder	Cat#LVS (VB181217-1124dqm)-C	Lentiviral construct to transfect and express the shRNA.
transfected construct (human)	siRNA to Dynamin 2 (SMARTpool)	Dharmacon/Thermo Fisher Scientific	Huang et al., 2004	transfected construct (human)

biological sample (<i>Macaca fascicularis</i>)	Primary simian hepatocytes	SingHealth, Singapore		Freshly isolated from <i>Macaca fascicularis</i>
antibody	anti-3-PGDH (Rabbit polyclonal)	Millipore	Cat# ABS571, RRID:AB_2783876	IF(1:1000), WB (1:1000)
antibody	anti-CRALBP (Mouse monoclonal)	Abcam	Cat# ab15051, RRID:AB_2269474	IF(1:500)
antibody	anti-human CD235a-APC (Mouse monoclonal)	Thermo Fisher Scientific	Cat#: 17-9987-42; RRID:AB_2043823	FACS (5 ul per test)
recombinant DNA reagent	PLKO-GFP (plasmid)	This paper		GFP version of pLKO.1-Puro
recombinant DNA reagent	PLKO.1-Puro (plasmid)	Sigma-Aldrich	RRID: Addgene_10878	Pol III based shRNA backbone
sequence-based reagent	Gipc1_F	This paper	PCR primers	GGGAAAG GACAAAA GGAACCC
sequenced-based reagent	Gipc1_R	This paper	PCR primers	CAGGGCA TTTGCACC CCATGCC
sequenced-based reagent	siRNA: nontargetin control	Thermo Fisher	4390843	Silencer Select
peptide, recombinant protein	Streptavidin	Thermo Fisher	Cat. #: 434302	
commercial assay or kit	In-Fusion HD Cloning	Clontech	Clontech:639647	

chemical compound, drug	CBR-5884	Sigma Aldrich	SML1656	
software, algorithm	SPSS	SPSS	RRID:SCR_002865	
other	DAPI stain	Invitrogen	D1306	(1 µg/mL)

Reagents

The following antibodies were used: rabbit anti-SEPT7 (1:100 dilution, #18991, IBL), Alexa-488-conjugated anti-rabbit antibody (1:500, #10729174, ThermoFisher Scientific), Alexa-647-conjugated anti-rabbit antibody (1:500, #A27040, ThermoFisher Scientific). The following dyes were used: Alexa-647-conjugated phalloidin (1:500, #10656353, ThermoFisher Scientific), Hoechst (1:500, #H3570, ThermoFisher Scientific), 0.01% (w/v) congo red (#C6767, Sigma-Aldrich). The following drugs were used: 50 µg/mL aphidicolin (#A0781 Sigma-Aldrich), 50 µg/mL cycloheximide (#2112 Cell Signalling), 2 µg/mL nalidixic acid (#N8878 Sigma-Aldrich), and 100 µg/ml rifampicin (#R3501 Sigma-Aldrich).

Bacterial strains and culture conditions

The bacterial strains and plasmids described in this study are listed in **Sup. Table 1** [↗](#). *Shigella flexneri* 5a str. M90T mCherry was used throughout the manuscript unless indicated otherwise unless indicated otherwise. *S. flexneri* strains were grown in trypticase soy broth (TCS) agar containing congo red to select for red colonies, indicative of a functional T3SS. Conical polypropylene tubes (#CLS430828, Corning) containing 5 ml of TCS were inoculated with individual red colonies of *S. flexneri* and were grown ~16 h at 37 °C with shaking at 200 rpm. The following day, bacterial cultures were diluted in fresh prewarmed TCS (1:50 or 1:100 v/v), and cultured until an optical density (OD, measured at 600 nm) of 0.6.

Cloning

Primers used in this study were designed using NEBuilder Assembly Tool (<https://nebuilder.neb.com/#/>). To produce *S. flexneri* expressing AfaI and GFP, the pACAfaI plasmid was constructed (plasmids characteristics and primer sequences are located in **Supplementary Table 1** [↗](#)). Briefly, the *afaI* operon together with the *amp^R* gene was amplified from *S. flexneri afaI* strain using the primers *afaI-operon-fw* and *afaI-operon-rv*. The *ori p15A* was amplified from pAC-P_{lac} using the primers *p15A-ori-fw* and *p15A-ori-rv*. Both DNA fragments were assembled using the Gibson assembly to generate the plasmid pAC-AfaI-Amp^R. In a second step, the *Amp^R* gene from pAC-AfaI-Amp was substituted by the *Cm^R* gene and generate pAC-AfaI-Cm. This was achieved by amplification of the *Cm^R* gene from pNeae2 using the primers *CmR-fw* *CmR-rv* and amplification of the backbone of pAC-AfaI-Amp with primers *afaI-p15A-fw*, *afaI-p15A-rv*, followed by Gibson assembly. Gibson assembly was performed at 50 °C for 30 min using the HiFi DNA Assembly Master Mix (#E2621L, New England Biolabs). Resulting plasmid pAC-AfaI-Cm was transformed into *S. flexneri* GFP by electroporation.

Cell lines

HeLa (ATCC CCL-2) cells were grown at 37 °C and 5% CO₂ in Dulbecco's modified Eagle medium (DMEM, GIBCO) supplemented with 10% fetal bovine serum (FBS, Sigma-Aldrich). Cells lines were tested for mycoplasma infection and tested negative.

Infection of human cells

Hela cells were seeded in 96-well microplates (PhenoPlates, Perkin Elmer) at a confluency of 10⁴ cells/well 2 days prior to infection. Alternatively, 9 × 10⁴ HeLa cells were seeded in 6-well plates (Thermo Scientific) containing 22 × 22 mm glass coverslips 2 days before the infection. Bacterial cultures were grown as described and cell cultures were infected with *S. flexneri* strains as described previously (Mostowy et al., 2010 [↗](#)). Briefly, Hela cells were infected with *S. flexneri* by spin-inoculation at 110 g for 10 min at a multiplicity of infection (MOI, bacteria:cell) of 100:1. Then, microplates were placed at 37 °C and 5% CO₂ for 30 min. Infected cultures were washed 3 × with phosphate-buffered saline (PBS) pH 7.4 and incubated with fresh DMEM containing 10% FBS and 50 mg/mL gentamicin at 37 °C and 5% CO₂ for 3 or 4 h. In order to obtain more instances of *S.*

Bacterial strain	Genotype	Reference
Shigella flexneri srv. 5a str. M90T	Sm ^R	(Mostowy et al., 2010)
<i>S. flexneri</i> mCherry	Constitutively producing mCherry, Carb ^R , Sm ^R	(Mostowy et al., 2013)
<i>S. flexneri</i> <i>afaI</i>	Constitutively producing the adhesin AfaE, Amp ^R , Sm ^R	(Mostowy et al., 2010)
<i>S. flexneri</i> GFP	Constitutively producing GFP, Carb ^R	(Mostowy et al., 2010)
<i>S. flexneri</i> GFP <i>afaI</i>	Constitutively producing the adhesin AfaE and GFP, Amp ^R Cm ^R , Sm ^R	This study
Plasmid	Characteristics	Reference
pTSAR1ud2.4s	T3SS reporter	(Campbell-Valois et al., 2014)
pAC-P _{lac}	pACYC184 derived plasmid.	(Lobato-Márquez et al., 2015)
pAC-AfaI-Amp ^R	For constitutive expression of the adhesin AfaE, ori p15A, Amp ^R	This study
pAC-AfaI-Cm ^R	For constitutive expression of the adhesin AfaE, ori p15A, Cm ^R	This study
Primer	Sequence (5'-3')	Reference
<i>afaI</i> -operon-fw	tgcgttgcgcgaagatccttgatctttctacgggg	This study
<i>afaI</i> -operon-rv	ggaagctaaacgagcccgatcttccca	This study
<i>p15A</i> -ori-fw	atcgggctcgtttagcttccttagctcc	This study
<i>p15A</i> -ori-rv	aaggatcttcgcgcaacgaattaatgtaag	This study
<i>afaI</i> - <i>p15A</i> -fw	cggggcgtaactgtcagaccaagtttac	This study
<i>afaI</i> - <i>p15A</i> -rv	tttctccatactcttcttttcaatatttg	This study
<i>CmR</i> -fw	aaggaagagtatggagaaaaaatcactggatataccaccgtgatatatccc	This study
<i>CmR</i> -rv	ggtctgacagttacgccccgcctgccca	This study

Supplementary table S1.

flexneri associated to SEPT7 in **Figure 4H** [↗](#), one of the datasets was obtained after infection with *S. flexneri* expressing the adhesin AfaI (Labigne-Roussel & Falkow, 1988 [↗](#)) and GFP to increase host cell invasion at a MOI of 10 in the absence of spin inoculation.

Click chemistry

To label *de novo* DNA and protein synthesis, Click-iT™ EdU Cell Proliferation Kit for Imaging (#C10337 ThermoFisher Scientific) and Click-iT™ AHA Protein Synthesis HCS Assay (#C10289 ThermoFisher Scientific) were used, respectively, according to the manufacturer instructions. In the case of AHA, cells were cultured in glutamine, cystine and methionine-free DMEM (#21013024 ThermoFisher Scientific) for 1 h prior to the treatment. 10μM EdU and 25μM AHA were supplemented to live bacterial or mammalian cell cultures for 1h. Cells were fixed for immunofluorescence as indicated below. Click azide/alkyne reactions were performed after permeabilization and prior to antibody staining as indicated below, according to the manufacturer's instructions.

Immunofluorescence and fluorescence microscopy

Bacteria, coverslips or 96-well microplates containing adherent infected or uninfected cells were washed three times with PBS pH 7.4 and fixed 15 min in 4% paraformaldehyde (in PBS) at RT. Fixed cells were washed three times with PBS pH 7.4 and subsequently permeabilized 5min with 0.1% Triton X-100 (in PBS). Cells were then washed three times in PBS and incubated 1h with primary antibodies diluted in PBS supplemented with 0.1% Triton X-100 and 1% bovine serum albumin. Cells were then washed three times in PBS and incubated 45 min with anti-rabbit secondary antibodies diluted 0.1% Triton X-100 (in PBS), and Hoechst and Alexa-conjugated phalloidin where indicated. 96-well microplates were washed three times in PBS and preserved in 0.01% sodium azide (#10338380 Thermo Scientific) in PBS pH 7.4. Stained bacteria cultures and coverslips were washed three times with PBS pH 7.4 mounted on glass slides with ProLong™ Diamond Antifade Mountant (#P36970I Invitrogen). Fluorescence microscopy on stained bacteria was performed using a using a 63×/1.4 C-Plan Apo oil immersion lens on a Zeiss LSM 880 confocal microscope driven by ZEN Blacksoftware (v2.3). Microscopy images were obtained using z-stack image series taking 8–16 slices. Fluorescence microscopy on infected or uninfected cells was performed using a ZEISS Plan-APOCHROMAT 20× / 0.95 Autocorr Objective or a ZEISS Plan-APOCHROMAT 50×/1.2 water immersion lens coupled to a 0.5x tubelens on a Zeiss CellDiscoverer 7 with Airyscan detectors driven by ZEN Blue software (v3.5). Microscopy images were obtained using z-stack image series taking 32 slices. Confocal images were processed using Airyscan processing (Weinerfilter) using “Auto Filter” and “3D Processing” options.

Flow cytometry

5×10^4 individual bacterial cells were analysed using flow cytometry with an LSRII flow cytometer (BD Biosciences). The data was analysed using FlowJo software, version 10.7.1. The median fluorescence of the bacterial population was determined and plotted for each staining.

Analysis pipelines

Number of bacteria per cell, as well as fluorescence of bacteria, nuclei and mammalian cells was analysed using CellProfiler (v.4.0.7) (Carpenter et al., 2006 [↗](#)). Custom made segmentation and deep learning pipelines for the classification of septin cages were written in ImageJ and Python using Keras.

Statistics

All graphs were plotted and statistical analysis were performed using Prism. n.s: non-significant, *: p-value < 0.05, **: p-value < 0.01, ***: p-value < 0.001).

Data availability

Due to the size of the data, representative datasets were deposited in Dryad (<https://doi.org/10.5061/dryad.6wwwpzgn5z>). Materials and additional raw microscopy data can be obtained from the corresponding authors upon requests. Custom ImageJ macros, python scripts and Jupyter notebooks were annotated and deposited in Github (<https://github.com/ATLopezJimenez/Toolset-high-content-analysis-of-Shigella-infection>).

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Additional files

Supplementary Figures

Additional information

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Reviewer #2 (Public review):

Summary:

Septin caging has emerged as one of the innate immune response of eukaryotic cells to infections by intracellular bacteria. This fascinating assembly of eukaryotic proteins into complex structures restricts bacteria motility within the cytoplasm of host cells, thereby facilitating recognition by cytosolic sensors and components of the autophagy machinery. Given the different types of septin caging that have been described thus far, a single cell, unbiased approach to quantify and characterise septin recruitment at bacteria is important to fully grasp the role and function of caging. Thus, the authors have developed an automated image analysis pipeline allowing bacterial segmentation and classification of septin cages

that will be very useful in the future, applied to study the role of host and bacterial factors, compare different bacterial strains or even compare infections by clinical isolates.

Strengths:

The authors developed a solid pipeline that has been thoroughly validated. When tested on infected cells, automated analysis corroborated previous observations and allowed the unbiased quantification of the different types of septin cages as well as the correlation between caging and bacterial metabolic activity. This approach will prove an essential asset in the further characterisation of septin cages for future studies.

Weaknesses:

As the main aim of the manuscript is to describe the newly developed analysis pipeline, the results illustrated in the manuscript are essentially descriptive. The developed pipeline seems exceptionally efficient in recognising septin cages in infected cells but its application for a broader purpose or field of study remains limited.

<https://doi.org/10.7554/eLife.97495.2.sa2>

Reviewer #4 (Public review):

Summary

In this study, López-Jiménez and colleagues demonstrate the utility of using high-content microscopy in dissecting host and bacterial determinants that play a role in the establishment of infection using *Shigella flexneri* as a model. The manuscript nicely identifies that infection with *Shigella* results in a block to DNA replication and protein synthesis. At the same time, the host responds, in part, via the entrapment of *Shigella* in septin cages.

Strengths:

The main strength of this manuscript is its technical aspects. They nicely demonstrate how an automated microscopy pipeline coupled with artificial intelligence can be used to gain new insights regarding elements of bacterial pathogenesis, using *Shigella flexneri* as a model system. Using this pipeline enabled the investigators to enhance the field's general understanding regarding the role of septin cages in responding to invading *Shigella*. This platform should be of interest to those who study a variety of intracellular microbial pathogens.

Another strength of the manuscript is the demonstration - using cell biology-based approaches - that infection with *Shigella* blocks DNA replication and protein synthesis. These observations nicely dovetail with the prior findings of other groups. Nevertheless, their clever click-chemistry-based approaches provide visual evidence of these phenomena and should interest many.

Weaknesses:

There are two main weaknesses of this work. First, the studies are limited to findings obtained using a single immortalized cell line. It is appreciated that HeLa cells serve as an excellent model for studying aspects of *Shigella* pathogenesis and host responses. However, it would be nice to see that similar observations are observed with an epithelial cell line of intestinal, preferably colonic origin, and eventually, with a non-immortalized cell line, although it is appreciated that the latter studies are beyond the scope of this work.

The other weakness is that the studies are minimally mechanistic. For example, the investigators have data to suggest that infection with *Shigella* leads to an arrest in DNA

replication and protein synthesis; however, no follow-up studies have been conducted to determine how these host cell processes are disabled. Interestingly, Zhang and colleagues recently identified that the *Shigella* OspC effectors target eukaryotic translation initiation factor 3 to block host cell translation (PMID: 38368608).

<https://doi.org/10.7554/eLife.97495.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public Review):

Summary:

*In this study, López-Jiménez and colleagues demonstrated the utility of using high-content microscopy in dissecting host and bacterial determinants that play a role in the establishment of infection using *Shigella flexneri* as a model. The manuscript nicely identifies that infection with *Shigella* results in a block to DNA replication and protein synthesis. At the same time, the host responds, in part, via the entrapment of *Shigella* in septin cages.*

Strengths:

*The main strength of this manuscript is its technical aspects. They nicely demonstrate how an automated microscopy pipeline coupled with artificial intelligence can be used to gain new insights regarding elements of bacterial pathogenesis, using *Shigella flexneri* as a model system. Using this pipeline enabled the investigators to enhance the field's general understanding regarding the role of septin cages in responding to invading *Shigella*. This platform should be of interest to those who study a variety of intracellular microbial pathogens.*

*Another strength of the manuscript is the demonstration - using cell biology-based approaches- that infection with *Shigella* blocks DNA replication and protein synthesis. These observations nicely dovetail with the prior findings of other groups. Nevertheless, their clever click-chemistry-based approaches provide visual evidence of these phenomena and should interest many.*

We thank the Reviewer for their enthusiasm on technical aspects of this paper, regarding both the automated microscopy pipeline coupled with artificial intelligence and the click-chemistry based approaches to dissect DNA replication and protein synthesis by microscopy.

Weaknesses:

*There are two main weaknesses of this work. First, the studies are limited to findings obtained using a single immortalized cell line. It is appreciated that HeLa cells serve as an excellent model for studying aspects of *Shigella* pathogenesis and host responses. However, it would be nice to see that similar observations are observed with an epithelial cell line of intestinal, preferably colonic origin, and eventually, with a non-immortalized cell line, although it is appreciated that the latter studies are beyond the scope of this work.*

The immortalized cell line HeLa is widely regarded as a paradigm to study infection by *Shigella* and other intracellular pathogens. However, we agree that future studies beyond the

scope of this work should include other cell lines (eg. epithelial cells of colonic origin, macrophages, primary cells).

*The other weakness is that the studies are minimally mechanistic. For example, the investigators have data to suggest that infection with *Shigella* leads to an arrest in DNA replication and protein synthesis; however, no follow-up studies have been conducted to determine how these host cell processes are disabled. Interestingly, Zhang and colleagues recently identified that the *Shigella* OspC effectors target eukaryotic translation initiation factor 3 to block host cell translation (PMID: 38368608). This paper should be discussed and cited in the discussion.*

We appreciate the Reviewer's concern about the lack of follow up work on observations of host DNA and protein synthesis arrest upon *Shigella* infection, which will be the focus of future studies. We acknowledge the recent work of Zhang et al. (Cell Reports, 2024) considering their similar results on protein translation arrest, and this reference has been more fully discussed in the revised version of the manuscript.

Reviewer #2 (Public Review):

Summary:

Septin caging has emerged as one of the innate immune responses of eukaryotic cells to infections by intracellular bacteria. This fascinating assembly of eukaryotic proteins into complex structures restricts bacteria motility within the cytoplasm of host cells, thereby facilitating recognition by cytosolic sensors and components of the autophagy machinery. Given the different types of septin caging that have been described thus far, a single-cell, unbiased approach to quantify and characterise septin recruitment at bacteria is important to fully grasp the role and function of caging. Thus, the authors have developed an automated image analysis pipeline allowing bacterial segmentation and classification of septin cages that will be very useful in the future, applied to study the role of host and bacterial factors, compare different bacterial strains, or even compare infections by clinical isolates.

Strengths:

The authors developed a solid pipeline that has been thoroughly validated. When tested on infected cells, automated analysis corroborated previous observations and allowed the unbiased quantification of the different types of septin cages as well as the correlation between caging and bacterial metabolic activity. This approach will prove an essential asset in the further characterisation of septin cages for future studies.

We thank the Reviewer for their positive comments, and for highlighting the strength of our imaging and analysis pipeline to analyse *Shigella*-septin interactions.

Weaknesses:

As the main aim of the manuscript is to describe the newly developed analysis pipeline, the results illustrated in the manuscript are essentially descriptive. The developed pipeline seems exceptionally efficient in recognising septin cages in infected cells but its application for a broader purpose or field of study remains limited.

The main objective of this manuscript is the development of imaging and analysis tools to study *Shigella* infection, and in particular, *Shigella* interactions with the septin cytoskeleton. In future work we will provide more mechanistic insight with novel experiments and broader applicability, using different cell lines (in agreement with Reviewer 1), mutants or

clinical isolates of *Shigella* and different bacteria species (eg. *Listeria*, *Salmonella*, mycobacteria).

Reviewer #3 (Public Review):

Summary:

The manuscript uses high-content imaging and advanced image-analysis tools to monitor the infection of epithelial cells by Shigella. They perform some analysis on the state of the cells (through measurements of DNA and protein synthesis), and then they focus on differential recruitment of Sept7 to the bacteria. They link this recruitment with the activity of the bacterial T3SS, which is a very interesting discovery. Overall, I found numerous exciting elements in this manuscript, and I have a couple of reservations. Please see below for more details on my reservations. Nevertheless, I think that these issues can be addressed by the authors, and doing so will help to make it a convincing and interesting piece for the community working on intracellular pathogens. The authors should also carefully re-edit their manuscript to avoid overselling their data (see below for issues I see there). I would consider taking out the first figure and starting with Figure 3 (Figure 2 could be re-organized in the later parts)- that could help to make the flow of the manuscript better.

Strengths:

The high-content analysis including the innovative analytical workflows are very promising and could be used by a large number of scientists working on intracellular bacteria. The finding that Septins (through SEPT7) are differentially regulated through actively secreting bacteria is very exciting and can steer novel research directions.

We thank the Reviewer for their constructive feedback and excitement for our results, including our findings on T3SS activity and *Shigella*-septin interactions. In accordance with the Reviewer's comments, we avoid overselling our data in the revised version of the manuscript.

Weaknesses:

The manuscript makes a connection between two research lines (1: Shigella infection and DNA/protein synthesis, 2: regulation of septins around invading Shigella) that are not fully developed - this makes it sometimes difficult to understand the take-home messages of the authors.

We agree that the manuscript is mostly technical and therefore some of our experimental observations would benefit from follow up mechanistic studies in the future. We highlight our vision for broader applicability in response to weaknesses raised by Reviewer 2.

It is not clear whether the analysis that was done on projected images actually reflects the phenotypes of the original 3D data. This issue needs to be carefully addressed.

We agree with the Reviewer that characterizing 3D data using 2D projected images has limitations.

We observe an increase in cell and nuclear surface that does not strictly imply a change in volume. This is why we measure Hoechst intensity in the nucleus using SUM-projection (as it can be used as a proxy of DNA content of the cell). However, we agree that future use of other markers (such as fluorescently labelled histones) would make our conclusions more robust.

Regarding the different orientation of intracellular bacteria, we agree that investigation of septin recruitment is more challenging when bacteria are placed perpendicular to the acquisition plane. In a first step, we trained a Convolutional Neural Network (CNN) using 2D data, as it is easier/faster to train and requires fewer annotated images. In doing so, we already managed to correctly identify 80% of *Shigella* interacting with septins, which enabled us to observe higher T3SS activity in this population. In future studies, we will maximize the 3D potential of our data and retrain a CNN that will allow more precise identification of *Shigella*-septin interactions and in depth characterization of volumetric parameters.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) To conclude that cell volume is indeed increased, the investigators should consider staining the cells with markers that demarcate cell boundaries and/or are confined to the cytosol, i.e., a cell tracker dye.

Staining using our SEPT7 antibody enables us to define cell boundaries for cellular area measurements (Novel Figure 1 - figure supplement 1A). However, we agree with the Reviewer that staining cells with additional markers (such as a cell tracker dye) would be required to conclude that cell volume is increased. We therefore adjust our claims in the main text (lines 107-115 and 235-246).

(2) Line 27: I understand what is meant by "recruited to actively pathogenic bacteria with increased T3SS activation." However, one could argue that there are many different roles of the intracytosolic bacteria in pathogenesis in terms of pathogenesis, not just actively secreting effectors.

T3SS secretion by cytosolic bacteria is tightly regulated and both T3SS states (active, inactive) likely contribute to the pathogenic lifestyle of *S. flexneri*. In agreement with this, we removed this statement from the manuscript (lines 27, 225 and 274).

(3) Line 88: Please clarify in the text that HeLa cells are being studied.

We explicitly mention that the epithelial cell line we study is HeLa in the main text (line 93), in addition to the Materials and methods (line 328).

(4) Line 97: is it possible to quantify the average distance of the nuclei from the cell perimeter? This would help provide some context as to what it means to be a certain distance from the nucleus, i.e., is there another way to point out that distance from nuclei correlates with movement inward post-invasion at the periphery?

To provide more context to the inward movement of bacteria to the cell centre, we provide calculations based on measurements in Figure 1G, I. If we approximate geometric shape of both cells and nucleus to a circle, the median radius of a HeLa cell is $31.1 \mu\text{m}^2$ (uninfected cell) and $36.3 \mu\text{m}^2$ (infected cell). Similarly, the median radius of the nucleus is $22.2 \mu\text{m}^2$ (uninfected cell) and $24.57 \mu\text{m}^2$ (infected cell).

However, we note that Figure 1F shows distance of bacteria to the centroid of the cell, which is the geometric centre of the cell, and which does not necessarily coincide with the geometric centre of the nucleus. We also note that nuclear area increases with infection (in a bacterial dose dependent manner). Finally, we note that these measurements are performed on max projections of 3D Z-stacks. In this case we cannot fully appreciate distance to the nucleus for bacteria located above it.

(5) Lines 212-213 - there is no Figure 9A, B - I think this should be Figure 7A, B.

Text has been updated (lines 216-217).

Reviewer #2 (Recommendations For The Authors):

Testing the analysis pipeline as a proof-of-concept question such as the comparison of caging around the laboratory strain as compared to one or a few clinical isolates or mutants of interest would help stress the relevance of this new, remarkable tool.

We thank the Reviewer for their enthusiasm.

Future research in the Mostowy lab will capitalise on the high-content tools generated here to explore the frequency and heterogeneity of septin cage entrapment for a wide variety of *S. flexneri* mutants and *Shigella* clinical isolates.

The sentence in line 215 ends with "in agreement with" followed by a reference.

Text has been updated (line 219).

The sentence in line 217 on the correlation between caging and T3SS is not very clear.

Text has been clarified (lines 221-223).

There is a typo in line 219 : "protrusSions"

Text has been updated (line 223).

Reviewer #3 (Recommendations For The Authors):

Major points

*The quantitative analysis approach in Figure 1 has multiple issues. Some examples:
(1) How was the cell area estimated? Normally, a marker for the whole cell (CellMask or similar) or cells expressing GFP would be good indicators. Here it is not clear to me what was done.*

The cell area was estimated using SEPT7 antibody staining which is enriched under the cell cortex. CellProfiler was used to segment cells based on SEPT7 staining, using a propagation method from the identified nucleus based on Otsu thresholding. To provide more clarity on how this was performed, we now include a new figure (Figure 1- figure supplement 1A) showing a representative image of HeLa cells stained with SEPT7 and the corresponding cell segmentation performed with CellProfiler software, together with an updated figure legend explaining the procedure (lines 784–787).

(2) The authors use Hoechst and integrated z-projections (Figure 1 S1) as a proxy to estimate nuclear volume. Hoechst staining depends on the organization of the DNA within the nucleus and I find that the authors need to do better controls to estimate nuclear size - this would be possible with cells expressing fluorescently labeled histones, or even better with a fluorescently tagged nuclear pore/envelope marker. The current quantification approach is misleading.

We understand Reviewer #3's concerns about using Hoechst staining as a proxy of nuclear volume, due to potential differences in DNA organisation within the nucleus.

Following the recommendation of Reviewer #3 in the following point 3, text has been updated (lines 107–115 and 235–246).

(3) Was cell density assessed for the measurements? If cells are confluent, bacteria could spread between cells within 3 hrs, if cells are less dense, this does not occur. When epithelial cells are infected for some hours, they have the tendency to round up a bit (and to appear thicker in z), but a bit smaller in xy. My suggestion to the authors (as they use these findings to follow up with experiments on the underlying processes) would be to tone down their statements - eg, Hoechst staining could be simply indicated as altered, but not put in a context of size (this would require substantial control experiments).

Local cell density was not directly measured, but the experiment was set up to infect at roughly 80% confluency (cells were seeded at 10^4 cells/well 2 days prior to infection in a 96-well microplate, as described in the Materials and methods section) and to ensure bacterial spread between cells.

In agreement with Reviewer #3 we tone down statements in the main text (see response to point 2 above).

In addition, I found Figure 1 (and parts of Figure 2) disconnected from the rest of the manuscript, and it may even be an idea to take it out of the manuscript (that could also help to deal with my feedback relating to Figure 1). I would suggest starting the manuscript with the current Figure 3 and building the biological story with a stronger focus on SEPT7 (and its links with T3 secretion and actively pathogenic bacteria) from there on. As it stands, the two parts of the manuscript are not well connected.

We carefully considered this comment but following revisions we have not reorganised the manuscript. We believe that high-content characterisation of *S. flexneri* infection in Figure 1 and 2 provides insightful information about changes in host cells in response to infection. Following this, we move onto characterising intracellular bacteria (and in particular those entrapped in septin cages) in the second part of the manuscript (Figure 3–7). Similar methods were used to analyse both host and bacterial cells and results obtained offer complementary views on host-pathogen interactions.

My major reservation with the experimental work of the current version of the manuscript relates to Figure 5: The analysis of the septin phenotypes in Figure 5 seems to be problematic - to me, it appears that analysis and training were done on projected image stacks. As bacteria are rod-shaped their orientation in space has an enormous impact on how the septin signal appears in a projection - this can lead to wrong interpretation of the phenotypes. The authors need to do some quantitative controls analyzing their data in 3D. To be more clear: the example "tight" (second row) shows a bacterium that appears short. It may be that it's actually longer if one looks in 3D, and the septin signal could possibly fall in the category "rings" or even "two poles".

The deep learning training and subsequent analysis of septin-cage entrapment is done on projected Z-stacks, which presents limitations. Future work in the Mostowy lab will exploit this first study and dive deeper into 3D aspects of the data.

To address Reviewer #3's concern, we include a sentence explaining that this analysis was performed using 2D max projections (lines 708 and 724), as well as acknowledging its limitations in the main text (lines 259–262).

Minor points

The scale bar in Fig 1 is very thin.

We corrected the scale bar in Fig. 1 to make it more visible.

Could it be that Figure 1F is swapped with Figure 1E in the description?

Descriptions for Figure 1E and F are correct.

Line 27: what does "actively pathogenic bacteria" mean? I propose to change the term.

We agree with Reviewer #3 that “actively pathogenic bacteria” should be removed from the text. This update is also in agreement with Reviewer #1 (see Reviewer #1 point 2).

Line 28: "dynamics" can be confusing as it relates to dynamic events imaged by time-lapse.

Although we are making a snapshot of the infection process at 3 hpi, we capture asynchronous processes in both host and bacterial cells (eg. host cells infected with different bacterial loads, bacterial cells undergoing actin polymerisation or septin cage entrapment). We agree that we are not following dynamics of full events over time. However, our high content approach enables us to capture different stages of dynamic processes. To avoid confusion, we replace “dynamics” by “diverse interactions” (line 28), and we discuss the importance of follow-up studies studying microscopy timelapses (line 274).

Paragraph 59 following: the concept of heterogeneity was investigated in some detail for viral infection by the Pelkmans group (PMID: 19710653) using advanced image analysis tools. Advanced machine-learning-based analysis was then performed on Salmonella invasion by Voznica and colleagues (PMID: 29084895). It would be great to include these somewhat "old" works here as they really paved the way for high-content imaging, and the way analyses were performed then should be also discussed in light of how analyses can be performed now with the approaches developed by the authors.

We agree. These landmark studies have now been included in the main text (lines 71-74).

Line 181: I do not know what "morphological conformations" means, perhaps the authors can change the wording or clarify.

We substituted the phrase “morphological conformations” by “morphological patterns” to improve clarity in the main text (lines 185).

The authors claim (eg in the abstract) that they are measuring the dynamic infection process. To me, it appears that they look at one time-point, so no dynamic information can be extracted. I suggest that the authors tone down their claims.

Please note our response above (Minor points, Line 28) which also refers to this question.

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